

GOOGLE PLAY STORE APP ANALYSIS AND RATING PREDICTION USING MACHINE LEARNING

A Data Science Project

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INTRODUCTION

The Google Play Store contains thousands of mobile apps with different categories, ratings, reviews, installations, sizes, and prices.

In this project, I analyzed Google Play Store data to understand different app features and patterns. I also built a Linear Regression model to predict the rating of a new app.

PROJECT FLOW

Google Play Store Dataset



Data Exploration



Data Cleaning



Outlier Removal



Data Visualization



Data Preprocessing



Linear Regression Model



Model Evaluation



New App Rating Prediction

DATA SET OVERVIEW

Dataset:

Google Play Store Dataset

It contains information about Android apps.Includes app details such as:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varies with device	4.2 and up
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1.1	4.4 and up

DATA EXPLORATION

Dataset Inspection= 1: df.info()

```
<class 'pandas.DataFrame'>
RangeIndex: 10841 entries, 0 to 10840
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   App                    10841 non-null  str    
1   Category               10841 non-null  str    
2   Rating                 9367 non-null   float64
3   Reviews                10841 non-null  str    
4   Size                   10841 non-null  str    
5   Installs               10841 non-null  str    
6   Type                   10840 non-null  str    
7   Price                  10841 non-null  str    
8   Content Rating         10840 non-null  str    
9   Genres                 10841 non-null  str    
10  Last Updated           10841 non-null  str    
11  Current Ver            10833 non-null  str    
12  Android Ver            10838 non-null  str    
dtypes: float64(1), str(12)
memory usage: 1.1 MB
```

2: df.describe()

Rating	
count	9367.000000
mean	4.193338
std	0.537431
min	1.000000
25%	4.000000
50%	4.300000
75%	4.500000
max	19.000000

DATA CLEANING

Handling Missing Values

- Removed records with missing **Rating** values.
- Replaced missing **Type** values with the most frequent value.
- Replaced missing **Content Rating** values with the most frequent value.

Removing Duplicate Records

- Checked the dataset for duplicate app entries.
- Removed duplicate records based on the **App** column.

Purpose

To improve data quality and ensure that the dataset was ready for further analysis and model building.

FEATURE CONVERSION

Preparing Numerical Features

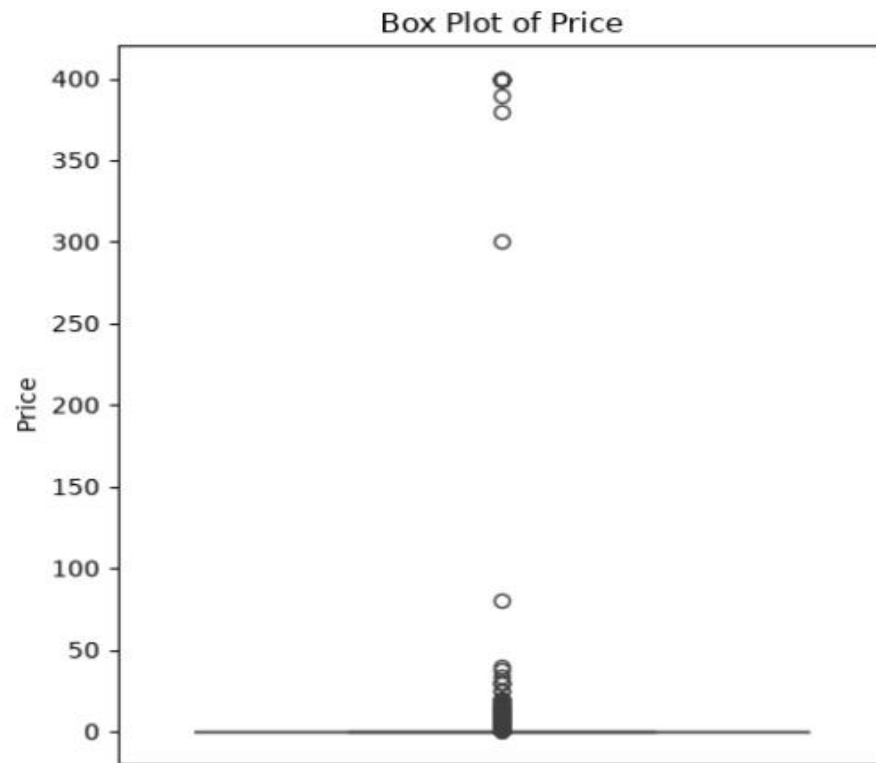
- Reviews column: Removed corrupted values and converted the column into numeric data.
- Installs column: Removed `+` and `,`, then converted the values into numbers.
- Price column: Removed the `\$` symbol and converted the values into numeric data.
- Size column: Converted values from MB and KB into a consistent format.
- Size column: Replaced “Varies with device” with missing values.

Purpose

To remove invalid values and convert inconsistent features into a clean, usable format for analysis and machine learning.

OUTLIER DETECTION AND REMOVAL

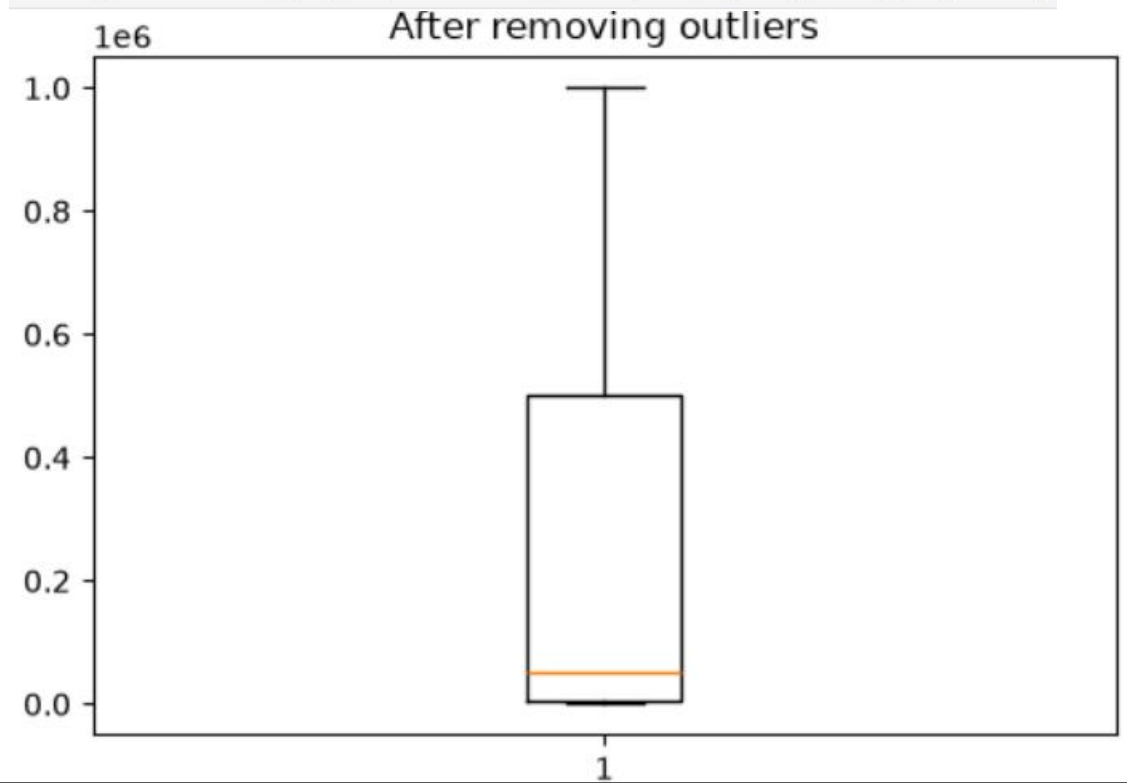
Boxplot: A boxplot shows the distribution and spread of numerical data while helping identify outliers.
An outlier is a data value that is unusually far away from most other values.



```
# Listing column
columns_to_clean = ['Price', 'Reviews', 'Installs']

# 2. Run a loop to calculate IQR and remove outliers for each column
for col in columns_to_clean:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outlier = df[(df[col] < lower_bound) | (df[col] > upper_bound)]

# Filter the dataframe to keep only values inside the boundaries
df_clean = df[(df[col] >= lower_bound) & (df[col] <= upper_bound)]
```



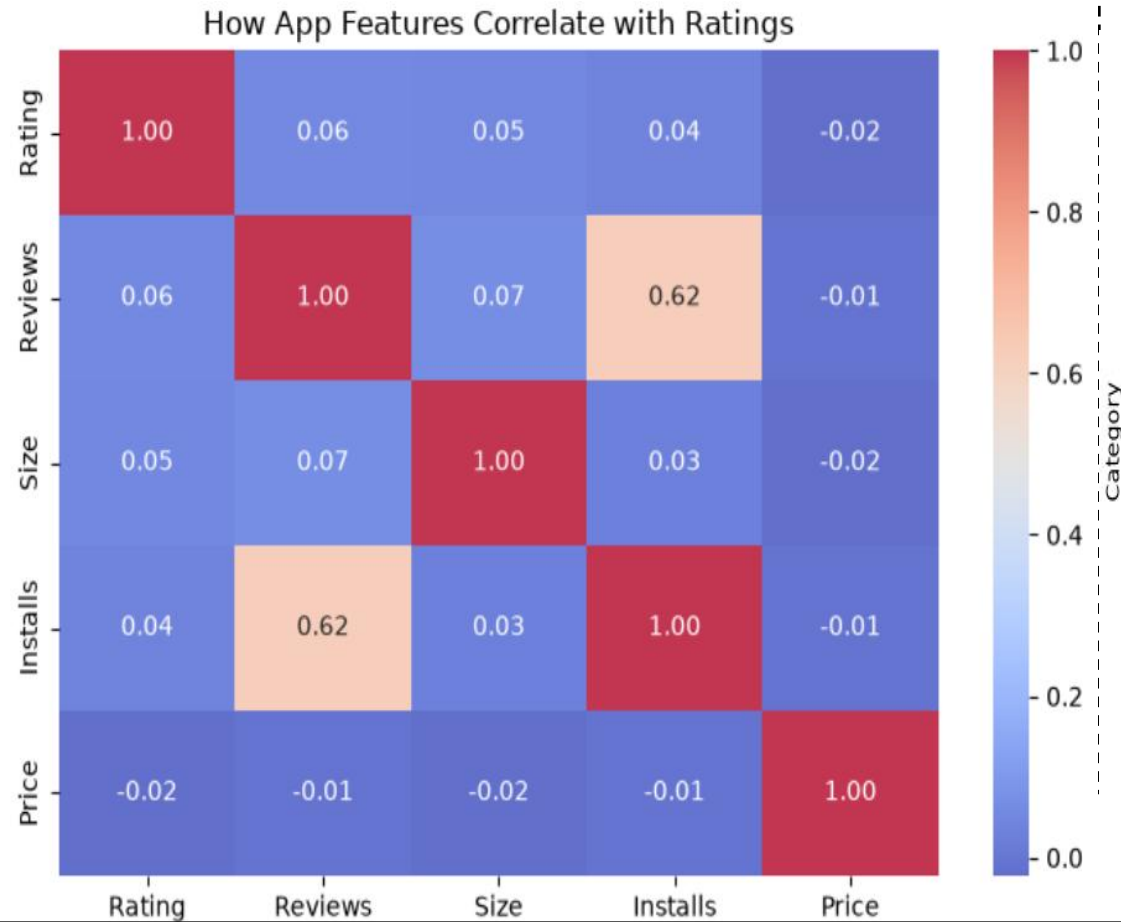
DATA ANALYSIS

Correlation Heatmap: Used to visualize the strength and direction of relationships between numerical variables. The correlation value ranges from -1 to +1:

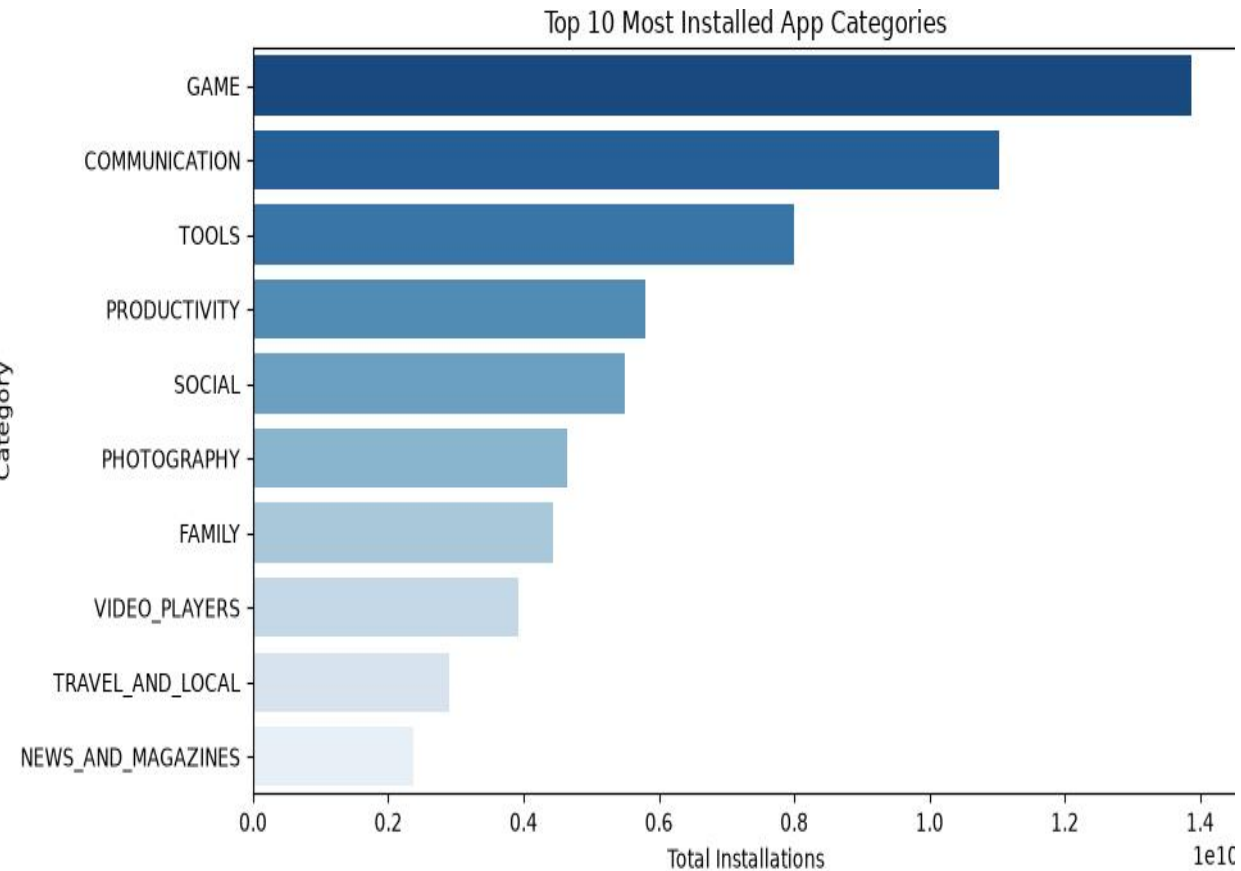
+1 → strong positive relationship

0 → little/no linear relationship

-1 → strong negative relationship



Bar Graph: Used to compare values across different categories and identify which categories are more or less common



DATA PREPROCESSING

Machine learning is a branch of AI where machines learn patterns from data using algorithms and use them to make predictions or decisions.

Machine learning models only understand numbers, not text. We have to convert our text columns into numbers using three different methods depending on what the data represents.

In this project i did three type of encoding depending on columns:

- 1. Label Encoding** (For "Type"):Replaces text with simple numbers (e.g., Free = 0, Paid = 1).
- 2. Custom Mapping** (For "Content Rating")Manually ranks text in a specific order (e.g., Everyone = 0 → Adults Only = 4).
- 3. One-Hot Encoding** (For "Category")Creates separate checklist columns of 0s and 1s (e.g., a "Sports" column gets a 1 if true, 0 if false).

Dropped First Column: Removed one column to prevent redundant data so the computer doesn't get confused by extra clues.

WHAT IS LINEAR REGRESSION

Supervised Learning

A type of ML where the model learns from labeled data, containing both inputs and their known outputs. Types of supervised learning:

Regression: Used to predict a continuous numerical value.

Example: Predicting house price.

Classification: Used to predict a category or class.

Example: IT Student / Non-IT Student.

Linear Regression is a supervised learning algorithm used to predict a numerical value by finding the relationship between one or more input variables and an output variable.

Seperate:

Input Features (X): The independent variables we cleaned and encoded to feed into the model: Reviews, Size, Installs, Price, Type, Content Rating, and Category columns.

Target Variable (y): The single dependent numerical value we want the model to predict: Rating (The app's final star score).

BUILDING THE MODEL

The 4-Step Process

Inputs (X) → 80/20 Data Split → Model Training → Prediction

The Data Roles

Target (y): The real app ratings we want to predict.

Inputs (X): The clean columns we use to make the prediction .

The 80/20 Data

Split 80% Training Data (X_train, y_train): The large part of the data used for the model to study and learn the patterns .

20% Testing Data (X_test, y_test): The small part kept hidden to act as a "final exam" to test the model .

Training and Testing

model.fit(): The model trains on the 80% data to find the best-fit line .

model.predict(): The model tests its skills on the hidden 20% data to make its final rating predictions (y_pred)

MAE, MSE, RMSE AND R² SCORE

1. MAE (Mean Absolute Error)

Simple Meaning: On average, how far is my prediction from the actual rating?

Low Value: Good. It means predictions are very close to the true ratings.

Large Value: Bad. It means model is guessing far away from the real numbers.

2. MSE (Mean Squared Error) & RMSE (Root Mean Squared Error)

Simple Meaning: What is the typical prediction error expressed in actual rating points.

Low Value: Good. It proves the model makes small, stable errors and cleaned out the outliers.

Large Value: Bad. It warns that the model is making massive, wild guessing mistakes.

3. R² Score (R-Squared)

Simple Meaning: How well does our model explain the variation and patterns in app ratings?

Low Value (Close to 0): The model cannot find a straight-line pattern. It indicates human ratings are too complex or subjective for a basic linear line.

Large Value (Close to 1): Excellent linear pattern. It indicates our input features explain the changes in the target rating almost perfectly.

MODEL RESULTS AND PREDICTION

Our Final Model Scores

MAE: 0.3885 (On average, our guesses are off by only 0.39 stars)

MSE: 0.3070 (Proves our errors are small and outliers are gone)

RMSE: 0.5541 (Our typical error stays within half a star).

R² Score: 0.0029 (Shows human ratings are too subjective for a straight line).

Testing a Brand-New App(game)

Reviews: 1,500

user reviews.Size: 25.0 Megabytes (MB).

Installs: 100,000 installations.

Price = 0.0: This designates that the application is completely Free to download.

Type = 0: This represents the Free classification

Content Rating = 0: This represents the baseline Everyone category

Final Prediction: 4.17 Stars

CONCLUSION

Data Cleaned: Bad data and outliers are gone.

Data Analyzed: Found patterns in app categories and ratings.

Model Trained: Built a Linear Regression model.

Model Tested: Evaluated errors using MAE, MSE, RMSE, and R^2 .

New App Predicted: Successfully predicted 4.17 Stars for a new app.

THANK YOU

